

A04 - Deep Learning for an 11-Year-Old Team Reflections

Mary Ann:

The nature of ideas sometimes produces unexpected outcomes that take on a life of their own. I did not set out to create a game. I had an idea and thought that maybe Claude AI could code my idea of how to teach an idea to an 11-year-old with a theoretical game. I provided the conditions and liked the initial game that was coded and updated the game several times before it was ready. The game is published on the internet so that it is accessible on the internet for anyone to use.

If I had to do this again, I would plan the game's rules and conditions to a far more thorough level than I had – it was a whim that the game could be coded with a prompt, and Claude came through. I used more time and compute resources chasing a whim than if I had properly developed the idea first before going to Claude to vibe code it. I used the free Claude version, and it coded the game and all the revisions within the daily allotted computer resources.

Cesar:

This project helped me understand entropy and cross-entropy better. At first, these concepts were difficult for me because they were related to probability and prediction. The biggest challenge was finding a simple way to explain them to an 11-year-old without using complicated formulas. The Dragon Egg Cave game helped me understand how uncertainty can change when we receive more information, and how cross-entropy can show if a prediction is close to the correct answer. I liked the creative part of the project because we used a game instead of only explaining the theory. This project showed me that learning Deep Learning is not only about understanding the technical concepts, but also about being able to explain them in a simple and clear way.

Luiz:

Working on this project helped me understand entropy and cross-entropy better because I was able to see the concepts through the Dragon Egg Cave game instead of only reading about them. The game made the ideas more visual and easier to understand. In the entropy section, I could see how having more eggs creates more uncertainty, and how clues about which egg is warmer can reduce that uncertainty. The cross-entropy part was also helpful because I could compare the human guess with the Dragon-Bot guess. The human has to choose one egg with 100% confidence, while the Dragon-Bot can spread its confidence between different eggs. After the correct egg is revealed, the game shows how the confidence of the prediction affects the result.

This helped me understand why being very confident in the wrong answer creates a bigger penalty. One thing I learned from this project is that explaining an AI concept to an 11-year-old can actually make you understand the concept better yourself. We had to think about how to make entropy and cross-entropy simple without changing their meaning. I liked that the game allowed the user to interact with the probabilities and see the results immediately.

Overall, this project helped me understand the practical meaning of entropy and cross-entropy and showed me how interactive examples can make difficult deep-learning concepts easier to learn. It also taught me that explaining a technical topic clearly is an important part of understanding it.

Nichelle:

This project made me think about the difference between understanding a technical concept and explaining it clearly to an 11-year-old. Our group chose entropy and cross-entropy and used an interactive dragon egg game to make the ideas more visual and engaging.

I thought the game looked creative and polished, but I initially had trouble understanding exactly how some of the mechanics were teaching the concepts. That had me question whether an 11-year-old might also become confused. As I worked through it more carefully, the ideas became clearer. Entropy was represented as uncertainty: when all eggs seemed equally likely, uncertainty was high, and when clues made one egg more likely, entropy decreased.

Cross-entropy was more challenging for me. I learned that it is not about how “close” a wrong answer is to the correct one, but how much probability or confidence was assigned to the answer that turned out to be correct. The Human vs. Dragon-Bot comparison also helped me understand how a model can distribute confidence across several possible classes instead of making only one absolute guess.

My biggest takeaway was how important wording is when simplifying technical material. Making something easier to understand should not change its meaning. This project helped me better understand entropy and cross-entropy while also showing me how much thought is required to explain deep learning concepts clearly to a younger audience.